

Comparativa de Semánticas Formales (NS vs SOS)

1. Semántica Natural (NS) – Rama 1 de 3

$$\mathcal{T}_1 = \frac{\langle x := 0, \sigma_1 \rangle \longrightarrow \sigma_2 \text{ [ass}_{\text{ns}}]}{\langle x := 0 \text{ or } (x := 1 \text{ or } x := 99), \sigma_1 \rangle \longrightarrow \sigma_2 \text{ [or1}_{\text{ns}}]}}$$

$\Downarrow$

$$\mathcal{T}_2 = \frac{\langle x := 5, \emptyset \rangle \longrightarrow \sigma_1 \text{ [ass}_{\text{ns}}]}{\langle S_1, \emptyset \rangle \longrightarrow \sigma_2} \mathcal{T}_1 \text{ [comp}_{\text{ns}}]}$$

Sentencias Abreviadas

$$S_1 = (x := 5 ; x := 0 \text{ or } (x := 1 \text{ or } x := 99))$$

Estados Registrados

$$\sigma_1 = \{x \mapsto 5\}$$

$$\sigma_2 = \{x \mapsto 0\}$$

1. Semántica Natural (NS) – Rama 2 de 3

$$\mathcal{T}_1 = \frac{\langle x := 1, \sigma_1 \rangle \longrightarrow \sigma_2 \text{ [ass}_{\text{ns}}]}{\langle x := 1 \text{ or } x := 99, \sigma_1 \rangle \longrightarrow \sigma_2} \text{ [or1}_{\text{ns}}]$$

$\Downarrow$

$$\mathcal{T}_2 = \frac{\mathcal{T}_1}{\langle x := 0 \text{ or } (x := 1 \text{ or } x := 99), \sigma_1 \rangle \longrightarrow \sigma_2} \text{ [or2}_{\text{ns}}]$$

$\Downarrow$

$$\mathcal{T}_3 = \frac{\langle x := 5, \emptyset \rangle \longrightarrow \sigma_1 \text{ [ass}_{\text{ns}}]}{\langle S_1, \emptyset \rangle \longrightarrow \sigma_2} \mathcal{T}_2 \text{ [comp}_{\text{ns}}]$$

Sentencias Abreviadas

$$S_1 = (x := 5 ; x := 0 \text{ or } (x := 1 \text{ or } x := 99))$$

Estados Registrados

$$\sigma_1 = \{x \mapsto 5\}$$

$$\sigma_2 = \{x \mapsto 1\}$$

1. Semántica Natural (NS) – Rama 3 de 3

$$\mathcal{T}_1 = \frac{\langle x := 99, \sigma_1 \rangle \longrightarrow \sigma_2 \text{ [ass}_{\text{ns}}]}{\langle x := 1 \text{ or } x := 99, \sigma_1 \rangle \longrightarrow \sigma_2} \text{ [or2}_{\text{ns}}]$$

$\Downarrow$

$$\mathcal{T}_2 = \frac{\mathcal{T}_1}{\langle x := 0 \text{ or } (x := 1 \text{ or } x := 99), \sigma_1 \rangle \longrightarrow \sigma_2} \text{ [or2}_{\text{ns}}]$$

$\Downarrow$

$$\mathcal{T}_3 = \frac{\langle x := 5, \emptyset \rangle \longrightarrow \sigma_1 \text{ [ass}_{\text{ns}}]}{\langle S_1, \emptyset \rangle \longrightarrow \sigma_2} \frac{\mathcal{T}_2}{\text{[comp}_{\text{ns}}]}$$

Sentencias Abreviadas

$$S_1 = (x := 5 ; x := 0 \text{ or } (x := 1 \text{ or } x := 99))$$

Estados Registrados

$$\sigma_1 = \{x \mapsto 5\}$$

$$\sigma_2 = \{x \mapsto 99\}$$

2. Semántica SOS – Rama 1 de 3

$$\mathcal{T}_1 = \frac{\langle x := 5, \emptyset \rangle \Rightarrow \sigma_1 [\text{ass}_{\text{sos}}]}{\langle S_1, \emptyset \rangle \Rightarrow \langle x := 0 \text{ or } (x := 1 \text{ or } x := 99), \sigma_1 \rangle} [\text{comp}_{\text{sos}}^2]$$

$\Downarrow$

$$\mathcal{T}_2 = \langle x := 0 \text{ or } (x := 1 \text{ or } x := 99), \sigma_1 \rangle \Rightarrow \langle x := 0, \sigma_1 \rangle [\text{or1}_{\text{sos}}]$$

$\Downarrow$

$$\mathcal{T}_3 = \langle x := 0, \sigma_1 \rangle \Rightarrow \sigma_2 [\text{ass}_{\text{sos}}]$$

Sentencias Abreviadas

$$S_1 = (x := 5 ; x := 0 \text{ or } (x := 1 \text{ or } x := 99))$$

Estados Registrados

$$\sigma_1 = \{x \mapsto 5\}$$

$$\sigma_2 = \{x \mapsto 0\}$$

2. Semántica SOS – Rama 2 de 3

$$\mathcal{T}_1 = \frac{\langle x := 5, \emptyset \rangle \Rightarrow \sigma_1 [\text{ass}_{\text{sos}}]}{\langle S_1, \emptyset \rangle \Rightarrow \langle x := 0 \text{ or } (x := 1 \text{ or } x := 99), \sigma_1 \rangle} [\text{comp}_{\text{sos}}^2]$$

$\Downarrow$

$$\mathcal{T}_2 = \langle x := 0 \text{ or } (x := 1 \text{ or } x := 99), \sigma_1 \rangle \Rightarrow \langle x := 1 \text{ or } x := 99, \sigma_1 \rangle [\text{or2}_{\text{sos}}]$$

$\Downarrow$

$$\mathcal{T}_3 = \langle x := 1 \text{ or } x := 99, \sigma_1 \rangle \Rightarrow \langle x := 1, \sigma_1 \rangle [\text{or1}_{\text{sos}}]$$

$\Downarrow$

$$\mathcal{T}_4 = \langle x := 1, \sigma_1 \rangle \Rightarrow \sigma_2 [\text{ass}_{\text{sos}}]$$

Sentencias Abreviadas

$$S_1 = (x := 5 ; x := 0 \text{ or } (x := 1 \text{ or } x := 99))$$

Estados Registrados

$$\sigma_1 = \{x \mapsto 5\}$$

$$\sigma_2 = \{x \mapsto 1\}$$

2. Semántica SOS – Rama 3 de 3

$$\mathcal{T}_1 = \frac{\langle x := 5, \emptyset \rangle \Rightarrow \sigma_1 [\text{ass}_{\text{sos}}]}{\langle S_1, \emptyset \rangle \Rightarrow \langle x := 0 \text{ or } (x := 1 \text{ or } x := 99), \sigma_1 \rangle} [\text{comp}_{\text{sos}}^2]$$

$\Downarrow$

$$\mathcal{T}_2 = \langle x := 0 \text{ or } (x := 1 \text{ or } x := 99), \sigma_1 \rangle \Rightarrow \langle x := 1 \text{ or } x := 99, \sigma_1 \rangle [\text{or2}_{\text{sos}}]$$

$\Downarrow$

$$\mathcal{T}_3 = \langle x := 1 \text{ or } x := 99, \sigma_1 \rangle \Rightarrow \langle x := 99, \sigma_1 \rangle [\text{or2}_{\text{sos}}]$$

$\Downarrow$

$$\mathcal{T}_4 = \langle x := 99, \sigma_1 \rangle \Rightarrow \sigma_2 [\text{ass}_{\text{sos}}]$$

Sentencias Abreviadas

$$S_1 = (x := 5 ; x := 0 \text{ or } (x := 1 \text{ or } x := 99))$$

Estados Registrados

$$\sigma_1 = \{x \mapsto 5\}$$

$$\sigma_2 = \{x \mapsto 99\}$$